Interval Estimation — Part 1

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Topics covered

1. Confidence Intervals — General Concepts
2. Confidence Interval for the Population Mean, μ, of Normal Populations

Reference: Newbold, P., Carlson, W., & Thorne, B. — *Statistics for Business and Economics*, Global Ed.

1) Confidence Intervals — General Concepts

Confidence Intervals — General Concepts

Definition

Once a random sample has been drawn from the population, **interval estimation** yields an interval that, with a specified degree of confidence, contains the true (unknown) parameter (interval estimate for a parameter θ).

Confidence Intervals — General Concepts

Methodology for constructing a confidence interval:

- Find a “good” point estimator;
- Establish a **confidence level** (most common: 90%, 95%, and 99%);
- Know the **sample size**;
- Know the **sampling distribution** of the estimator.

Confidence Intervals — General Concepts

In choosing the estimator, the Pivotal Variable Method should be followed.

According to this method, the **pivot statistic** (or pivotal variable):

- Must contain the parameter to be estimated in its expression;
- Its sampling distribution (exact or approximate) must **not depend** on the parameter, nor on any other unknown quantity.

2) CI for the Population Mean, μ

Confidence Interval for μ — Overview

Case 1 — Normal population, σ **known**

Case 2 — Normal population, σ **unknown**

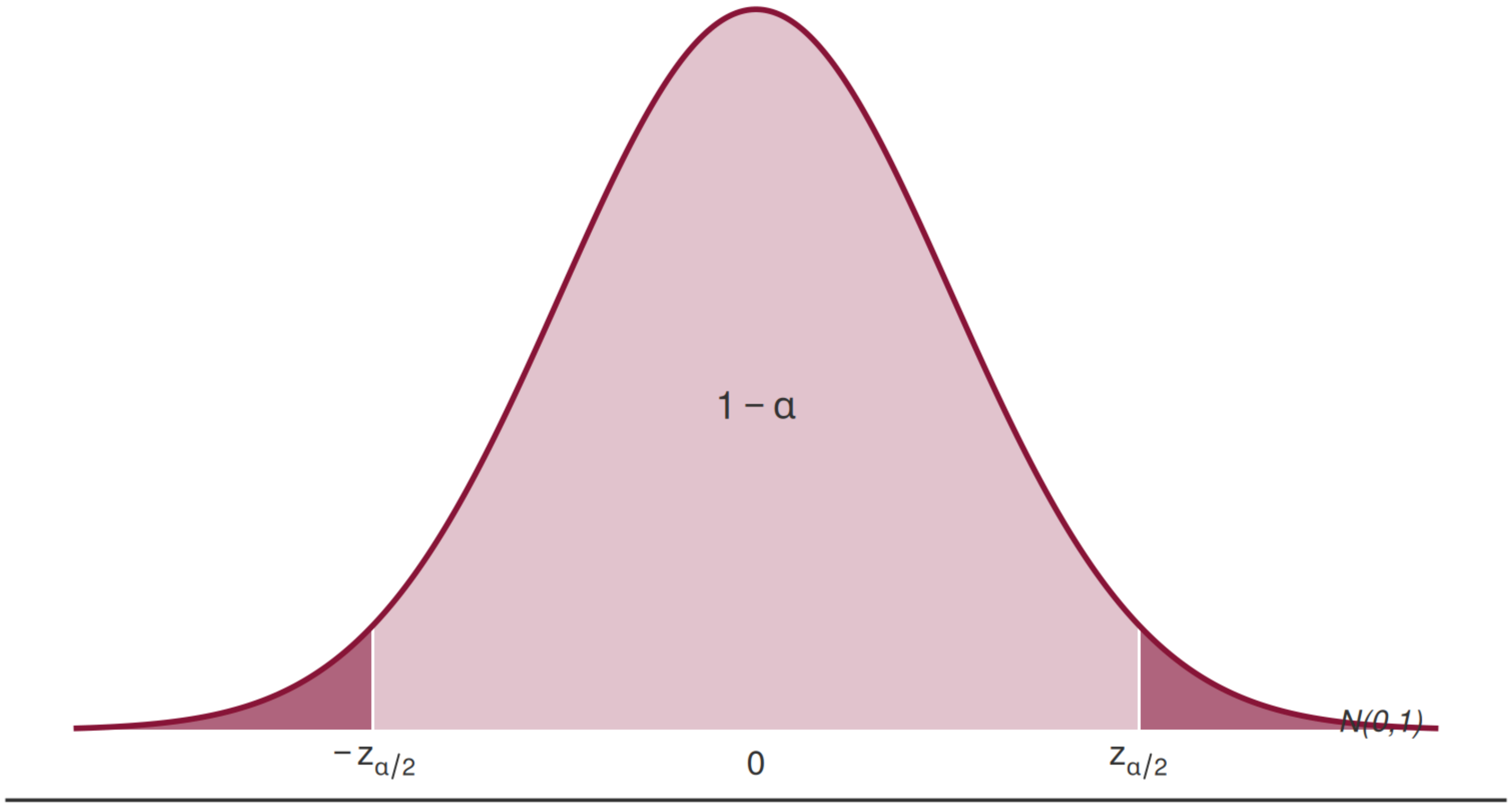
Case 1: Normal Population and σ Known

Case 1 — Normal Population and σ Known

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$, drawn from a population X with distribution $N(\mu, \sigma)$, where σ is **known**.

$$\underbrace{\mu}_{\text{parameter}} \longrightarrow \underbrace{\bar{X}}_{\substack{\text{point}\\\text{estimator}}} \longrightarrow \underbrace{Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)}_{\text{pivot statistic}}$$

Case 1 — Setting Up the Probability Statement



$$P\left(-z_{\alpha/2} < Z < z_{\alpha/2}\right) = 1 - \alpha$$

Confidence attributed to the interval: $(1 - \alpha) \times 100\%$

Case 1 — Deriving the Confidence Interval

$$P\left(-z_{\alpha/2} < Z < z_{\alpha/2}\right) = 1 - \alpha$$

$$\Updownarrow$$

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

$$\Updownarrow \quad \cdots$$

$$P\left(\underbrace{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{T_1} < \mu < \underbrace{\bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}_{T_2}\right) = 1 - \alpha$$

Case 1 — The Confidence Interval

The $(1-\alpha)\times 100\%$ **confidence interval for** μ is:

$$\boxed{IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)}$$

where $t_1 = \bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ and $t_2 = \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ are the **observed** lower and upper bounds.

Case 1 — Random Nature of the Bounds

Different samples yield different values of $\bar{x}$ and, consequently, different values of the bounds t_1 and t_2.

Therefore, those bounds are realizations of **random variables** T_1 and T_2, respectively.

> The confidence interval is **random** — it varies from sample to sample. What we compute from our data is one particular realization of that random interval.

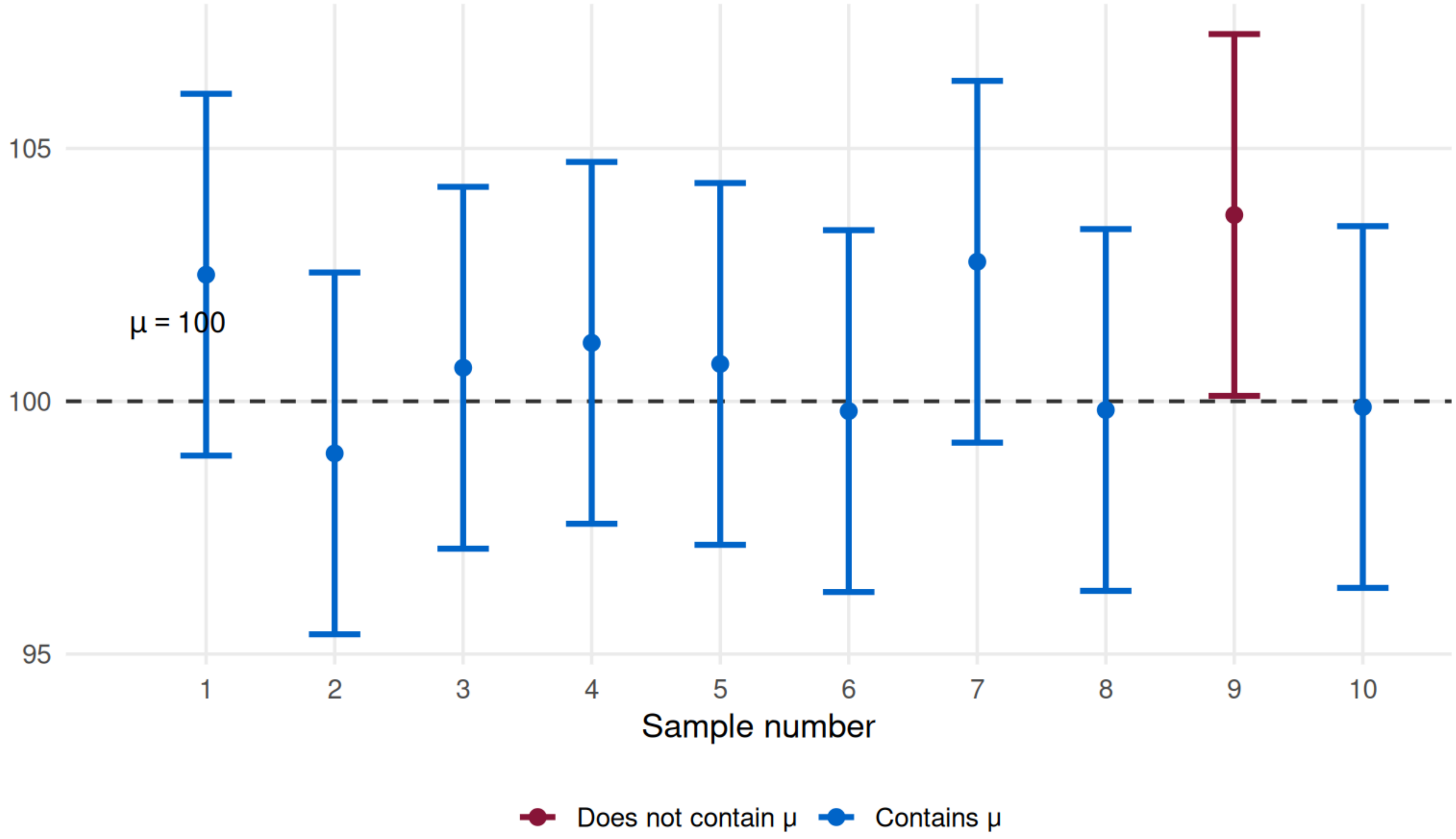
Case 1 — Interpretation

Correct Interpretation

If an **infinite number** of random samples of the same size were drawn, and a $(1-\alpha) \times 100\%$ confidence interval for μ were computed from each sample, then $(1-\alpha) \times 100\%$ **of those intervals** would contain the true value of μ.

A particular computed interval either contains μ or it does not — we just do not know which. The $(1-\alpha) \times 100\%$ refers to the **long-run coverage** of the procedure.

Case 1 — Visualization of Coverage



Case 1 — Properties of the CI

$$IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

The interval is **symmetric**: the midpoint equals the point estimate $\bar{x}$, and $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ is the standard error of the estimator $\bar{X}$.

The **estimation error** is the maximum error committed:

$$|\bar{X} - \mu| < z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \qquad \longleftarrow \text{estimation error}$$

Case 1 — Width of the Interval

$$IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- **Confidence level** $\uparrow \Rightarrow$ width **increases** $\Rightarrow$ inference becomes **less precise** (and vice versa).
- **Variance** $\uparrow \Rightarrow$ width **increases**, because the standard error of the estimator increases.
- **Sample size** $\uparrow \Rightarrow$ width **decreases** $\Rightarrow$ inference becomes **more precise**.

Case 1 — A Practical Note

$$IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

It is **not** guaranteed that constructing a CI always produces useful information. It is necessary to strike a balance between:

- the **sample size**,
- the **confidence level**, and
- the **precision** of the interval.

Case 1 — Application Exercise

Application Exercise — Setup

Consider a population with a Normal distribution and known standard deviation $\sigma = 20$.

A random sample of size $n = 20$ was drawn, yielding a sample mean $\bar{x} = 320$.

Find the 90% confidence interval for the population mean.

Application Exercise — Conditions (Case 1)

We are in the conditions of Case 1 (Normal population, σ known):

$$X \sim N(\mu,\ \sigma = 20), \qquad n = 20, \qquad \bar{x} = 320$$

The pivot statistic is:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

Application Exercise — Finding $z_{\alpha/2}$

The confidence interval to use is:

$$IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

We want 90% confidence $\Rightarrow 1-\alpha = 0.90 \Rightarrow \alpha = 0.10$.

We have $\bar{x}$, σ, and n. We still need $z_{\alpha/2}$.

For the most common confidence levels (90%, 95%, …), the values of $z_{\alpha/2}$ are tabulated (Table 5).

Looking at the table, for $\alpha = 0.10$:

$$z_{\alpha/2} = z_{0.05} = 1.645$$

Application Exercise — Result

Substituting all values into the CI formula:

$$IC_{90\%}(\mu) = \left(320 - 1.645 \times \frac{20}{\sqrt{20}},\ 320 + 1.645 \times \frac{20}{\sqrt{20}}\right)$$

90% Confidence Interval for μ

$$IC_{90\%}(\mu) = (312.64\ ;\ 327.36)$$

Application Exercise — Interpretation

$$IC_{90\%}(\mu) = (312.64\ ;\ 327.36)$$

The particular interval $(312.64\ ;\ 327.36)$ is a **90% confidence interval** for the true value of μ.

This means: if we were to construct confidence intervals from many different samples of the same size, **90% of them would effectively contain the population mean μ**.

Case 2: Normal Population and σ Unknown

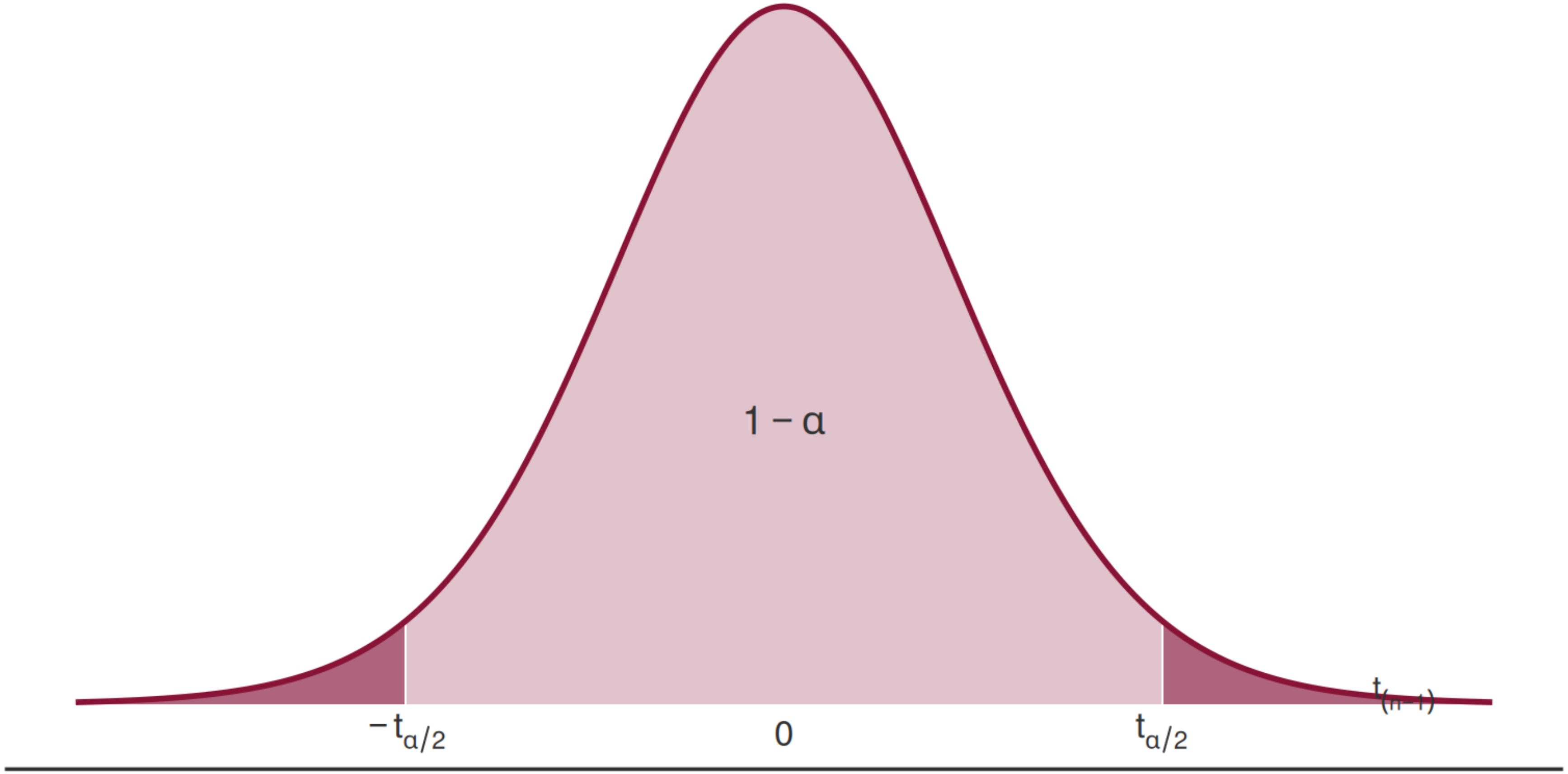
Case 2 — Normal Population and σ Unknown

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$, drawn from a population X with distribution $N(\mu, \sigma)$, where σ is **unknown**.

$$\underbrace{\mu}_{\text{parameter}} \longrightarrow \underbrace{\bar{X}}_{\substack{\text{point}\\ \text{estimator}}} \longrightarrow \underbrace{T = \frac{\bar{X} - \mu}{S'/\sqrt{n}} \sim t_{(n-1)}}_{\text{pivot statistic}}$$

where S' is the corrected sample standard deviation.

Case 2 — Setting Up the Probability Statement



$$P\left(-t_{\alpha/2} < T < t_{\alpha/2}\right) = 1 - \alpha \qquad \text{(confidence: } (1 - \alpha) \times 100\backslash\%\text{)}$$

Case 2 — The Confidence Interval

The $(1-\alpha)\times 100\%$ **confidence interval for** μ is:

$$\boxed{IC_{(1-\alpha)\times 100\%}(\mu) = \left(\bar{x} - t_{\alpha/2}\frac{s'}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}\frac{s'}{\sqrt{n}}\right)}$$

where $t_{\alpha/2}$ is the critical value from the $t_{(n-1)}$ distribution.

> All properties derived for the CI in Case 1 (see earlier slides) apply equally here: the interval is symmetric, and its width increases with the confidence level and variance, and decreases with the sample size.

Case 2 — Application Exercise 1

Exercise 1 — Setup

Based on a random sample of $n = 16$ observations from a **Normal population**, the following 90% confidence interval for the expected value was constructed:

$$(7.398\ ;\ 12.602)$$

Given that $s' = 3.872$, determine the **confidence level** that can be attributed to this interval.

(a) 98% (b) 99% (c) Between 98% and 99%

Exercise 1 — Conditions (Case 2)

Conditions: Case 2 — Normal population, σ unknown.

$$X \sim N(\mu, \sigma = ?\,), \qquad n = 16, \qquad s' = 3.872$$

Pivot statistic: $T = \dfrac{\bar{X} - \mu}{S' / \sqrt{n}} \sim t_{(n-1)} \equiv t_{(15)}$

Exercise 1 — Finding $t_{\alpha/2}$

The interval given is:

$$IC_{(1-\alpha)\times 100\%}(\mu) = (7.398\ ;\ 12.602)$$

Its **width** is:

$$h = 12.602 - 7.398 = 5.204$$

The width of the CI can also be expressed as:

$$h = 2 \times t_{\alpha/2} \frac{s'}{\sqrt{n}}$$

Exercise 1 — Solving for $t_{\alpha/2}$

Setting the two expressions for h equal:

$$2 \times t_{\alpha/2} \frac{s'}{\sqrt{n}} = 5.204 \;\Rightarrow\; t_{\alpha/2} = \frac{5.204 \times \sqrt{n}}{2 \times s'}$$

$$t_{\alpha/2} = \frac{5.204 \times \sqrt{16}}{2 \times 3.872} = \frac{5.204 \times 4}{7.744} = 2.688$$

From the t-Student table (Table 7, row for 15 d.f.):

$$\underset{\text{area}=0.01}{2.602} < 2.688 < \underset{\text{area}=0.005}{2.947}$$

Exercise 1 — Conclusion

$$\underbrace{2.602}_{\text{area}=0.01} < 2.688 < \underbrace{2.947}_{\text{area}=0.005}$$

$$0.005 < \frac{\alpha}{2} < 0.01 \iff 0.01 < \alpha < 0.02 \iff 0.98 < 1 - \alpha < 0.99$$

The confidence level of the given interval lies **between 98% and 99%**.

The correct answer is (c).

Case 2 (continued): σ Unknown and $n > 30$

Case 2 — Large Samples ($n > 30$)

Consider a random sample $X_1, X_2, \ldots, X_n$, $n \in \mathbb{N}$ and $n > 30$, from a population $X \sim N(\mu, \sigma)$ with σ **unknown**.

$$\underbrace{\mu}_{\text{parameter}} \longrightarrow \underbrace{\bar{X}}_{\substack{\text{point}\\ \text{estimator}}} \longrightarrow \underbrace{Z = \frac{\bar{X} - \mu}{S'/\sqrt{n}} \mathrel{\dot{\sim}} N(0,1)}_{\text{pivot statistic (approx.)}}$$

The pivot statistic **no longer follows** a $t_{(n-1)}$ distribution — it follows an **approximately** $N(0, 1)$ distribution.

Case 2 — Large-Sample CI ($n > 30$)

The **approximate** $(1-\alpha) \times 100\%$ confidence interval for μ is:

$$IC_{(1-\alpha)\times 100\%}(\mu) \approx \left(\bar{x} - z_{\alpha/2}\frac{s'}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{s'}{\sqrt{n}}\right)$$

Note: When $n > 30$ and σ is unknown, the corrected sample standard deviation s' is used in place of σ, and the **standard normal** critical value $z_{\alpha/2}$ replaces the $t_{\alpha/2}$ critical value. The resulting interval is **approximate**.

Case 2 — Application Exercise 2

Exercise 2 — Setup

Using the same sample from Exercise 1 ($n = 16$, $s' = 3.872$, CI at 98%: $(7.398\ ;\ 12.602)$):

What sample size should be collected (assuming s' does not change) **so that the width of the CI is reduced by half?**

Exercise 2 — Initial Analysis

Current width: $h = 5.204$

New target width: $h' = h/2 = 5.204/2 = 2.602$

Increasing precision while keeping all other conditions constant requires a **larger sample**. We assume $n > 30$, so the pivot becomes approximately $N(0, 1)$.

Exercise 2 — Solving for n

The approximate CI for μ becomes:

$$IC_{98\%}(\mu) \approx \left(\bar{x} - z_{\alpha/2}\frac{s'}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{s'}{\sqrt{n}}\right)$$

The new width satisfies:

$$h' = 2 \times z_{\alpha/2}\frac{s'}{\sqrt{n}} \Rightarrow 2.602 = 2 \times \underset{\text{Table 5, } \varepsilon=0.02}{\underset{\uparrow}{2.326}} \times \frac{3.872}{\sqrt{n}}$$

$$\sqrt{n} = \frac{2 \times 2.326 \times 3.872}{2.602} \approx 6.92$$

$$n \geq (6.92)^2 \Rightarrow \boxed{n = 48}$$

Exercise 2 — Conclusion

Required Sample Size

To reduce the width of the 98% CI by half (from 5.204 to 2.602), while keeping $s' = 3.872$ unchanged, a sample of $n = 48$ observations must be collected.

Note that the assumption $n > 30$ is satisfied ($48 > 30$), validating the use of the normal approximation.

Summary — Case 1 vs. Case 2

	Case 1	Case 2 ($n \leq 30$)	Case 2 ($n > 30$)
Population	$N(\mu, \sigma)$	$N(\mu, \sigma)$	$N(\mu, \sigma)$
σ	Known	Unknown	Unknown
Pivot	$Z \sim N(0, 1)$	$T \sim t_{(n-1)}$	$Z \dot{\sim} N(0, 1)$
Critical value	$z_{\alpha/2}$	$t_{\alpha/2}$	$z_{\alpha/2}$
SE	$\sigma/\sqrt{n}$	$s'/\sqrt{n}$	$s'/\sqrt{n}$
Interval	Exact	Exact	Approximate